



Debug

Training Assignment

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RECORD OF CHANGES

No	Effective Date	Change Description	Reason	Reviewer	Approver

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CODE	:	BAD_Assignment09_v1.0
TYPE	:	N/A
LOC	:	N/A
DURATION	:	60 minutes

Day 9: Debug

Assignment 1: Apply the knowledge learned in the debugging lesson to find and fix errors in the code below

```
class BinarySearch
```

```
fun main(args: Array<String>){  
    print(binarySearch(23, arrayOf(12, 3, 24, 5, 10, 23, 9)))  
}  
  
fun binarySearch(element: Int, array: Array<Int>): Int {  
    Arrays.sort(array)  
    var index: Int = 0  
    var end = array.size - 1  
    while(index <= end){  
        val center: Int = (index + end) / 2  
        if (element == array[center]){  
            return center  
        }else if (element < array[center]){  
            end = center - 1  
        }else if(element > array[center]){  
            index = center + 2  
        }  
    }  
    return -1  
}
```

Technical Requirements:

- Indicates detailed errors and how to correct them

Assignment 2: Apply the knowledge learned in the debugging lesson to find and fix errors in the code below

```
fun main(args: Array<String>) {  
    val a = 2.3  
    val b = 4  
    val c = 5.6  
    val root1: Double  
    val root2: Double  
    val output: String  
    val determinant = b * b - 4.0 * a * c  
    // condition for real and different roots  
    if (determinant > 0) {  
        root1 = (-b + Math.sqrt(determinant)) / (2 * a)  
        root2 = (b - Math.sqrt(determinant)) / (2 * a)  
  
        output = "root1 = %.2f and root2 = %.2f".format(root1, root2)  
    }  
    // Condition for real and equal roots  
    else if (determinant == 0.0) {  
        root2 = -b / (2 * a)  
        root1 = root2  
        output = "root1 = root2 = %.2f;".format(root1)  
    }  
    // If roots are not real  
    else {  
        val realPart = -b / (2 * a)  
        val imaginaryPart = Math.sqrt(determinant) / (2 * a)
```

```
        output = "root1 = %.2f+%.2fi and root2 = %.2f-%.2fi".format(realPart, imaginaryPart, realPart,
        imaginaryPart)
    }
    println(output)
}
```

Technical Requirements:

- Indicates detailed errors and how to correct them

Estimated Time to complete: 60p

-- THE END --